The Cox model with Breslow's ties, and its predicted survival

The Mayo Clinic model for primary biliary cirrhosis, reproduced from the 1989 paper.

Tested against	Dickson, Grambsch, Fleming, Fisher and Langworthy (1989), *Hepatology*: Table 2 (both models) and Table 4 (baseline survival, years 1–7). The 418 Mayo patients as Fleming and Harrington (1991) published them (PollisDatasets `pbc`).
Missing in Julia	Breslow's rule for tied deaths, and the predicted survival curve of a patient from a fitted Cox model. Survival.jl uses Efron's rule and does not predict survival curves.
Notebook	**Compose** › **New Methods** › **Browse Methods**: *The Cox model with Breslow's ties, and its predicted survival* (`cox.ipynb`)
Result	the initial model and all seven years of predicted survival to every printed digit; 41 of 53 values in all. With Survival.jl's Efron ties, only 1 to 3 decimals.

The problem

When several patients die on the same day, the partial likelihood of Cox (1972) needs a rule for the tie. Survival.jl uses the rule of Efron (1977). Much of the published literature, and the SAS, BMDP and early S software it was computed with, used that of Breslow (1974), so its coefficients cannot be reproduced in Julia. A model that cannot be reproduced cannot be checked, updated or applied to new patients with confidence. And a clinical model is used through its predictions: "what is this patient's chance of surviving five years?", which Survival.jl does not compute.

The method

Breslow's ties

All patients who die on the same day share one risk set: each death contributes $\eta_i - \log \sum_{j \in R(t_i)} e^{\eta_j}$, with $\eta = X\beta$. `cox` maximises it by Newton's method with exact derivatives from ForwardDiff, and returns coefficients, standard errors from the inverse information, Wald chi-squares and the likelihood ratio chi-square.

Predicted survival

A patient with risk score $R = X\beta$ has $S(t, R) = S_0(t)^{\exp(R-c)}$, c the mean risk score. The S_0 of Kalbfleisch and Prentice (1980) is a product, over the days with deaths, of factors α, each the root of $\sum_{\text{deaths}} w/(1-\alpha^w) = \sum_{\text{at risk}} w$ with $w = \exp(R-c)$; `survival` finds each root by bisection.

What the tests show

Test	What is compared	Agreement
Survival.jl	data without ties: Breslow is then Cox	6 decimals or more
Table 2, initial	312 trial patients: all 20 values	every printed digit
Table 2, final	418 patients	last digit in 6 values (see below)
Table 2	likelihood ratio chi-square	199.13 vs 199.2 (open)
Table 4	$S_0(t)$, years 1 to 7	every printed digit
Efron (Survival.jl)	Table 2, for contrast	only 1 to 3 decimals

The contrast in the last row is the point: on the same data, with the same covariates, the only difference between a fit that reproduces the paper and one that does not is the rule for ties.

The final model, in the last digit

Log bilirubin 0.8706 vs 0.8707, log prothrombin time 2.379 vs 2.380, and four chi-squares by 0.01 or 0.02. The paper estimated missing values for 16 declined patients with regressions fitted to the 90 complete ones (p. 3); the public file lacks only two prothrombin times, so the regression here is simpler. Setting those two times to anything from 9 to 13 seconds moves the coefficients in the third decimal, gaps of this size. The initial model, which needs no estimated values, agrees throughout.

The paper's two patients

A 52-year-old with bilirubin 0.5, albumin 4.5, prothrombin time 10.1 and no edema: the paper gives a 96% chance of

surviving 5 years. One with bilirubin 13.9, albumin 2.8, prothrombin time 13.8 and edema treated with diuretics: 43% of surviving 1 year. The notebook draws both curves from the final model.

Still open

- **The likelihood ratio chi-square**, 199.13 against 199.2, although every coefficient and standard error it comes from agrees. Not explained.
- **The one-sample log-rank test** the paper used to validate the model on 106 patients (p. 5) is not included: its groups use a biopsy variable that is not in the public data, and their sizes are not given, so there are no expected results to test against.

Further considerations

- **Ties matter more than they seem.** Survival data recorded in days, months or years are full of ties; the choice between Breslow and Efron is a convention of the software, and a reader of an older paper has to know which was used.
- **Upstream.** A `ties` option and a `predict` for survival curves in Survival.jl would make every Breslow-era model in the literature reproducible in Julia. Tables 2 and 4 are ready-made tests.
- **Next.** Time-varying covariates (the Mayo model was later updated for repeated visits), stratification, and confidence bands for $S(t, R)$, each with a published table to test it.

References

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Methods written and tested with Claude Code (Anthropic) in Pollis, from the published papers and their data.

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